REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 6

Station-Keeping and Orbit Maintenance

Holding an Orbit for a Whole Mission: Dead-bands, Reboosts and Delta-v Budgets

Prepared for Cao Khanh Tan

Module 6, in the Advanced Application Phase

Builds on Session 2 (perturbations) and Session 3 (delta-v, rocket equation)

Tools STK Astrogator and lifetime tools, GMAT, Python

SESSION 6

Station-Keeping and Orbit Maintenance

Session 2 explained why orbits drift: J2, drag, luni-solar gravity and solar radiation pressure all pull the elements away from their design values. This module is about fighting back. Station-keeping is the routine of small corrective burns that hold a satellite inside its allowed slot for years, and orbit maintenance is the wider job of managing altitude, ground track and end-of-life. It is where the perturbation physics of Session 2 meets the delta-v and rocket-equation tools of Session 3, and it decides how much propellant a mission must carry, which often decides how long it can live.

Learning objectives

- Explain the dead-band control cycle and how a maintenance manoeuvre works.
- Relate a small burn to a change in semi-major axis, altitude or inclination.
- Break down a geostationary station-keeping budget into North-South and East-West.
- Estimate the reboost delta-v a low orbit needs to fight atmospheric drag.
- Turn a lifetime delta-v budget into a propellant mass fraction, and compare chemical with electric.
- Describe end-of-life disposal: deorbit for low orbits, graveyard for geostationary.

Station-keeping in one picture

The controlled quantity drifts under perturbations until it reaches the edge of an allowed box, a small burn resets it, and the cycle repeats. The mission carries enough propellant for every cycle across its whole life. Everything below is either the size of one correction or the sum of them all.

PART 1

The Dead-band Control Cycle

A satellite is not held exactly on its nominal state; it is allowed to drift within a tolerance box, the dead-band. When the controlled quantity (a longitude slot for a GEO satellite, an altitude for a LEO one) reaches the edge, a small burn pushes it back to the other side, and it drifts across again. The wider the box, the fewer burns, but the looser the control.

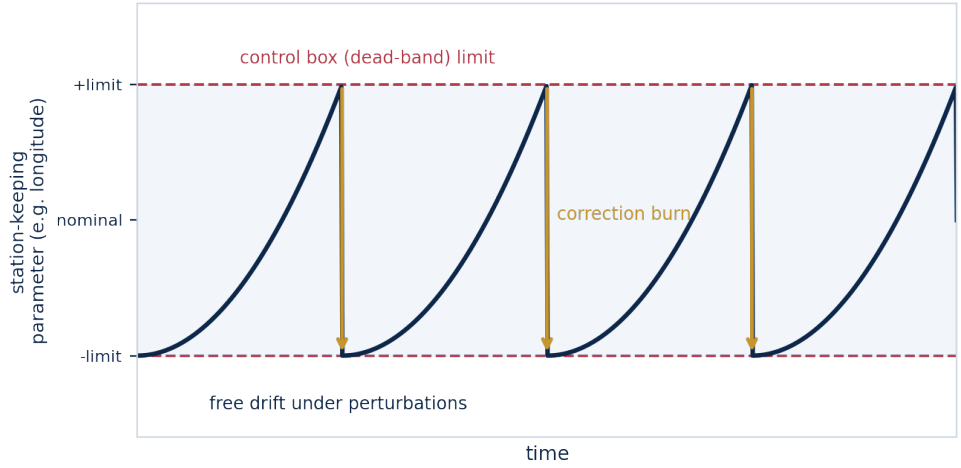


Figure 1. The dead-band cycle. The controlled parameter drifts under perturbations to the edge of the allowed box, a correction burn (gold) resets it, and the cycle repeats. The curvature reflects a near-constant perturbing acceleration, as in East-West longitude control.

Two small-manoeuvre relations do most of the work. A tangential burn changes the orbital energy, and therefore the semi-major axis (and altitude); a burn perpendicular to the orbit at a node rotates the plane, changing inclination:

$$\Delta v \approx \frac{n}{2}\,\Delta a$$

$$\Delta v = 2v\sin\frac{\Delta i}{2} \approx v\,\Delta i$$

where n is the mean motion and v the orbital speed. Both say the same thing: small corrections cost little delta-v, but they must be repeated for the life of the mission.

PART 2

Geostationary Station-Keeping

A geostationary satellite must be held in a small box in both latitude and longitude, typically a fraction of a degree on each side. The two directions are controlled separately and cost very different amounts.

North-South (inclination) keeping. Luni-solar gravity drives the inclination up by about 0.85 to 0.95 degrees per year. Correcting it is a plane-change burn at a node, so its yearly cost is the orbital speed times the annual inclination drift:

$$\dot{v}_{\mathrm{NS}} \approx v\,\dot{i} \quad \text{(per year)}$$

At the GEO speed of 3.07 km/s, that is roughly 46 to 51 metres per second each year, and it is by far the largest station-keeping cost.

East-West (longitude) keeping. The Earth is slightly triaxial, so a satellite off its stable longitude feels a tangential pull that changes its semi-major axis and makes it drift in longitude at a rate set by that error:

$$\dot{\lambda} \approx -\frac{3}{2}\,\omega_{\oplus}\,\frac{\Delta a}{a}$$

Correcting it costs only about 2 to 4 metres per second per year. The two together give the classic geostationary budget:

Component	Fights	Annual cost
North-South (inclination)	Luni-solar gravity	about 45 to 51 m/s per year
East-West (longitude)	Earth triaxiality and SRP	about 2 to 4 m/s per year
Total		roughly 50 m/s per year

PART 3

Low-Orbit Maintenance: Fighting Drag

In low orbit the dominant enemy is atmospheric drag, which continuously removes energy and lowers the semi-major axis. The decay per revolution grows with air density and with the ballistic coefficient:

$$\Delta a_{\text{rev}} = -2\pi \frac{C_D A}{m} \rho a^2$$

Left alone, the orbit spirals down and re-enters. To maintain altitude, the satellite periodically reboosts, a tangential burn that raises the semi-major axis back up. Each reboost costs, from the small-manoeuvre relation above:

$$\Delta v \approx \frac{n}{2} \Delta a$$

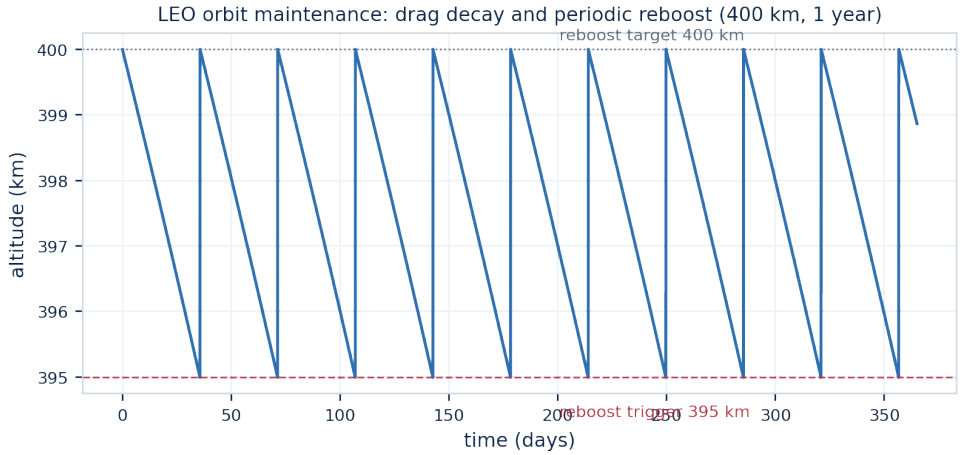


Figure 2. One year of drag decay and reboost for a 400 km orbit. The altitude sags a few kilometres under drag, then a reboost restores it. The sawtooth repeats for the life of the mission; the ISS is maintained exactly this way.

Worked example: a 400 km reboost budget

A satellite at 400 km with a typical ballistic coefficient decays by a few kilometres a month. Holding it within a 5 km band takes reboosts totalling about 28 m/s per year, rising sharply when solar activity heats and expands the atmosphere. Lower orbits are cheaper to launch to but far more expensive to maintain, and below about 300 km the drag cost becomes prohibitive.

PART 4

Lifetime Budgets and Propulsion Choice

Station-keeping is a yearly cost, so the mission budget is the annual rate multiplied by the design life:

$$\Delta v_{\text{SK}} = \left(\dot{v}_{\text{NS}} + \dot{v}_{\text{EW}}\right) t_{\text{life}}$$

A 15-year geostationary mission at 53 m/s per year needs about 795 m/s of station-keeping delta-v, on top of whatever it spent reaching orbit. That total converts to propellant through the rocket equation from Session 3:

$$\frac{m_p}{m_0} = 1 - e^{-\Delta v/(I_{sp}\, g_0)}$$

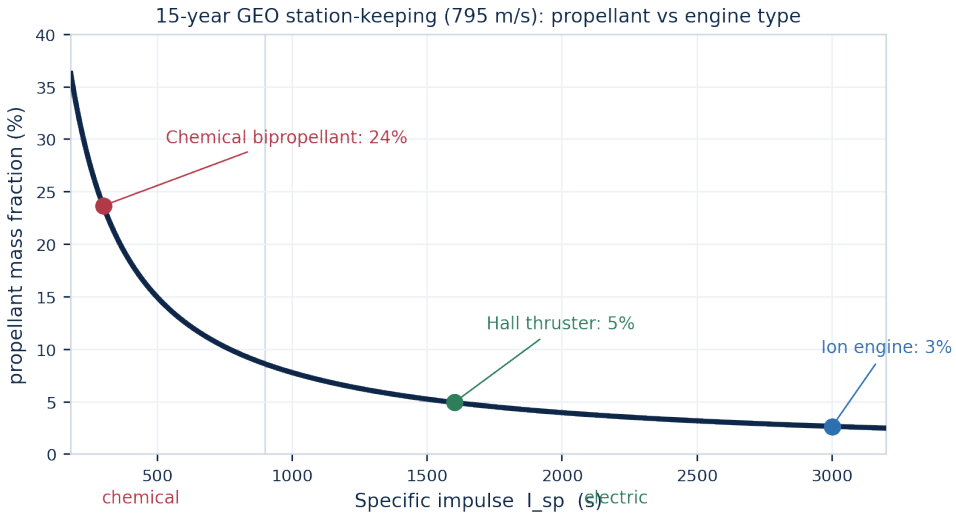


Figure 3. The same 795 m/s budget costs very different amounts of propellant depending on the engine. A chemical bipropellant system burns about a quarter of the satellite mass; a Hall or ion thruster needs only a few percent, because its specific impulse is five to ten times higher.

This is why almost every modern geostationary satellite now does its station-keeping, and increasingly its orbit-raising, with electric propulsion. The penalty is that low-thrust manoeuvres take much longer, so the maintenance burns are spread over hours rather than delivered as sharp impulses.

PART 5

End of Life: Deorbit and Graveyard

Orbit maintenance ends with disposal, now a regulatory requirement to limit debris. The strategy depends on the orbit:

- **Low orbits: deorbit.** A retrograde burn lowers the perigee into the atmosphere so drag finishes the job; guidelines now call for re-entry within about five years of end of mission. Very low orbits deorbit naturally.
- **Geostationary: graveyard.** There is nowhere to deorbit to, so the satellite is raised a few hundred kilometres above the ring into a disposal orbit, clearing the operational belt.

The graveyard raise is a small semi-major axis change, so it is cheap:

$$\Delta v_{\text{grave}} = \frac{n}{2}\,\Delta a_{\text{grave}}$$

Worked example: the graveyard burn
Raising a geostationary satellite 300 km above the ring costs delta-v = (n/2) x 300 km, which is only about 11 m/s, roughly three months of station-keeping. Operators reserve this small amount at end of life; a satellite that runs out of propellant before disposal becomes a hazard, which is why the lifetime budget always keeps the graveyard burn in hand.

Computing Maintenance Budgets

The whole module is a few lines: annual rates times life for the budget, the rocket equation for propellant, and the small-manoeuvre relations for each correction.

```
import numpy as np
mu, g0 = 398600.4418, 9.80665
a_geo = 42164.0; n = np.sqrt(mu/a_geo**3)      # rad/s

# GEO lifetime station-keeping budget
ns, ew, life = 50.0, 3.0, 15                   # m/s per year, years
dv_sk = (ns + ew) * life                        # ~795 m/s
prop  = 1 - np.exp(-dv_sk/(300*g0))             # chemical Isp 300 s (dv in m/s)

# small-manoeuvre relations (n in rad/s, da in km -> dv in km/s)
dv_reboost = (n/2) * 5.0                          # raise SMA by 5 km
dv_grave   = (n/2) * 300.0                        # graveyard, +300 km
print(dv_sk, prop, dv_grave*1000, 'm/s')
```

Production tools

STK Astrogator scripts real station-keeping with targeted burns and dead-band logic, and STKs Lifetime tool estimates how long a low orbit survives against drag for a given ballistic coefficient and solar-activity model. GMAT does the same through its propagators and targeting sequences. The hand budget here is what tells you the answer is the right order of magnitude.

PRACTICE

This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

Assignment: Station-Keeping and Orbit Maintenance

- **Reading and notes (3 to 4 h).** Explain the dead-band control cycle and why North-South keeping dominates the geostationary budget while East-West is nearly free. One page.
- **GEO lifetime budget (3 to 4 h).** For a 12-year geostationary mission, compute the total station-keeping delta-v (take North-South 48 m/s/yr, East-West 3 m/s/yr), then the propellant mass fraction for a chemical stage (Isp 300 s) and an ion stage (Isp 3000 s). Add the graveyard burn.
- **LEO reboost (3 to 4 h).** Using the drag-decay model from the notes, estimate the reboost delta-v per year to hold a 450 km orbit within a 5 km band. Compare it to a 350 km orbit and comment on the trend.
- **Small manoeuvres (2 h).** Compute the delta-v to raise a GEO satellite 250 km into a graveyard orbit, and the delta-v to correct a 0.9 degree inclination error at GEO.
- **STK (2 h).** Use the STK Lifetime tool on a low orbit to estimate its decay time, or set up a simple Astrogator station-keeping sequence, and capture the result.

Answer key (for your self-check)

Quantity	Expected value
12-year GEO budget	(48+3) x 12 = 612 m/s
Propellant fraction, Isp 300 / 3000	about 0.19 / 0.021
450 km reboost	order 10 to 20 m/s per year (rises steeply lower down)
Graveyard raise, +250 km	delta-v = (n/2) x 250 km, about 9.1 m/s
0.9 deg inclination fix at GEO	delta-v = v x 0.9 deg, about 48 m/s

REFERENCE

One-Page Formula Sheet

Lifetime station-keeping budget	$\Delta v_{SK} = (\dot{v}_{NS} + \dot{v}_{EW})\, t_{life}$
North-South (inclination) rate	$\dot{v}_{NS} \approx v\,\dot{i}$ (per year)
SMA / altitude correction	$\Delta v \approx \frac{n}{2}\,\Delta a$
Inclination correction	$\Delta v = 2v\sin\frac{\Delta i}{2} \approx v\,\Delta i$
Longitude drift from SMA error	$\dot{\lambda} \approx -\frac{3}{2}\,\omega_{\oplus}\,\frac{\Delta a}{a}$
Drag decay per revolution	$\Delta a_{rev} = -2\pi\frac{C_D A}{m}\,\rho\, a^2$
Propellant mass fraction	$\frac{m_p}{m_0} = 1 - e^{-\Delta v/(I_{sp}\, g_0)}$
Graveyard raise	$\Delta v_{grave} = \frac{n}{2}\,\Delta a_{grave}$

Key glossary

- **Station-keeping:** the routine small burns that hold a satellite inside its allowed slot.
- **Dead-band:** the tolerance box the satellite is allowed to drift within before a correction.
- **North-South / East-West keeping:** control of inclination (costly) and longitude (cheap) at GEO.
- **Reboost:** a tangential burn that restores altitude lost to drag in low orbit.
- **Graveyard orbit:** a disposal orbit a few hundred km above the geostationary ring.

Going further

For depth, see Vallado, Fundamentals of Astrodynamics and Applications (station-keeping and maneuver planning), Wertz and Larson, Space Mission Analysis and Design (delta-v budgets and propulsion sizing), and Montenbruck and Gill, Satellite Orbits (drag and lifetime). These are real, widely used texts; please confirm editions yourself, as I cannot verify those details from here.

Coming up in Session 7
Rendezvous and proximity operations: relative motion in orbit, the Clohessy-Wiltshire equations, and how one spacecraft approaches and docks with another. The final building block before the capstone.